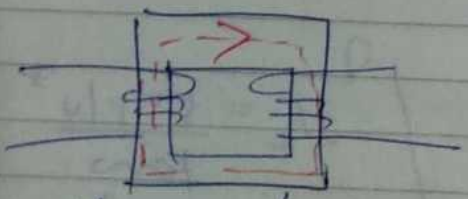
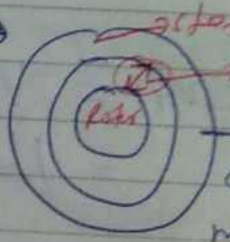
UNIT-II

⇒ 3-ϕ Induction motor

Transformer	Induction motor
→ It is a static device	→ It is a mechanical device which converts electrical energy into mechanical energy
→ It is an electromagnetic energy conversion device	→ It is an electro-mechanical energy conversion device
→ It is a constant frequency device	→ It is a variable frequency device ie $f = sf$
→  So, it is having continuous magnetic circuit. So, T/F draws less magnetizing current from supply. So, $I_\phi \rightarrow (4+6\%)\ I_f$	→  It is having discontinuous magnetic ckt. So, flux has to reach from stator to rotor by passing through air gap. Since airgap is more, reluctance is more, so, Induction motor draws more magnetizing current from supply. $I_\phi = (25 \text{ to } 35\%)\ I_f$

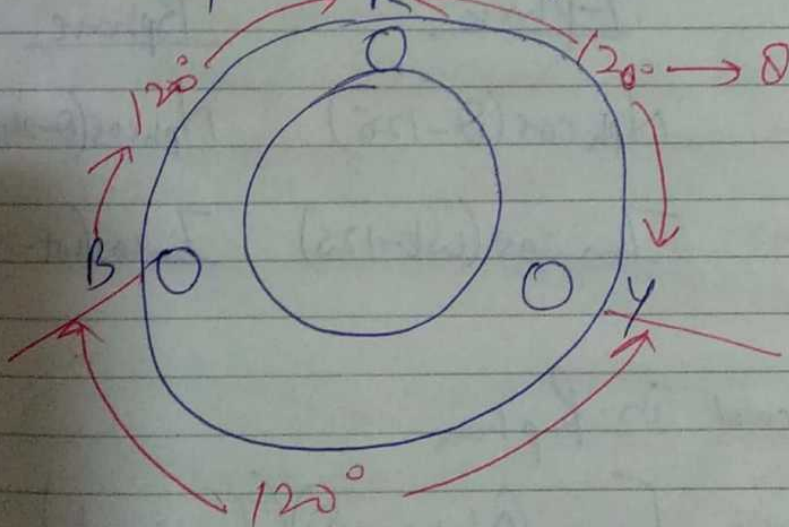
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So,
→ Efficiency is η↑ (because of less loss)
→ Efficiency is η↓ (because of more losses)

⇒ Production of RMF (Rotating magnetic field)

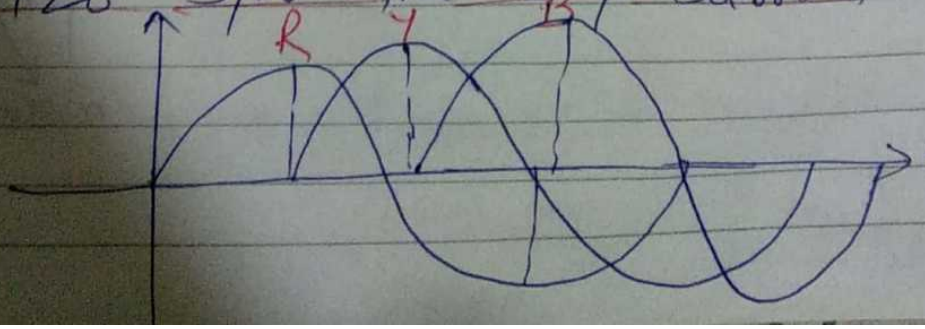
→ Conditions are :-

(1) There should be a space angle displacement of 120° b/w 3-phases



So, for any m-phase system
$\Rightarrow \frac{2\pi}{3} = 120°$

(2) There should be a time angle displacement of 120° b/w the 3-ϕ currents

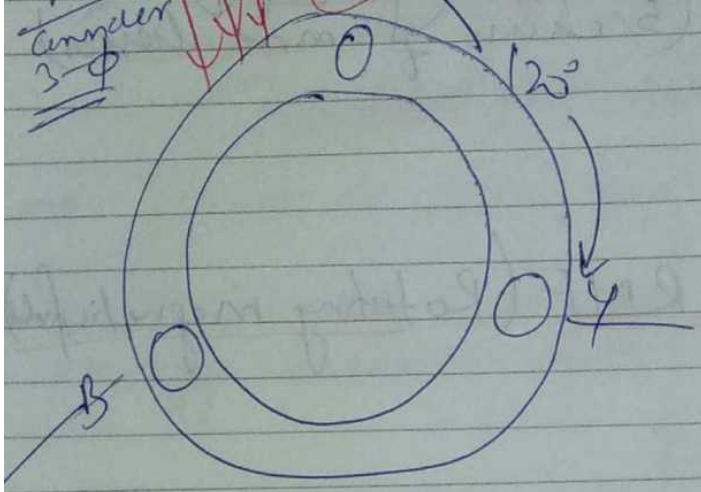


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Applying 3-ϕ supply to stator

Let current 3-ϕ

Consider R phase as reference



R-phase	Y-phase	B phase
$N_{ph}\cos\theta$	$N_{ph}\cos(\theta-120°)$	$N_{ph}\cos(\theta-240°)$
$I_m\cos\omega t$	$I_m\cos(\omega t-120°)$	$I_m\cos(\omega t-240°)$

So, mmf produced in R phase

→ i.e. $f_R = (N_{ph}\cos\theta)(I_m\cos\omega t)$

$f_R = N_{ph} I_m \cos\theta\cos\omega t$

$\therefore \boxed{f_R = f_m\cos\theta\cos\omega t}$

$\{f_m = N_{ph} I_m\}$

maximum mmf

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// ly mmf produced in Yphase

$$f_y = [N_{ph}\cos(\theta-120^\circ)][I_m\cos(\omega t-120^\circ)]$$

$$F_y = N_{ph} I_m \cos(\theta-120^\circ)\cos(\omega t-120^\circ)$$

(or)

$$\boxed{f_y = f_m \cos(\theta-120^\circ)\cos(\omega t-120^\circ)}$$

// ly mmf produced in Bphase

$$\boxed{F_B = f_m \cos(\theta-240^\circ)\cos(\omega t-240^\circ)}$$

&, The Resultant mmf in 3-ϕ m/c :-

$$\boxed{\bar{f}_{res} = \bar{f}_R + \bar{f}_Y + \bar{f}_B}$$

$$\boxed{\bar{f}_{res} = \frac{3}{2} f_m \cos(\theta-\omega t)}$$

$$\therefore \boxed{f_{res} = f_{max}\cdot\cos(\theta-\omega t)}$$

max resultant mmf in 3-ϕ m/c $\left\{ f_{max} = \frac{3}{2} f_m \right\}_{(N_{ph}\cdot I_m)}$

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& the speed of Rotating magnetic field is always Synchronous Speed ie

$$N_s = \frac{120f}{P}$$

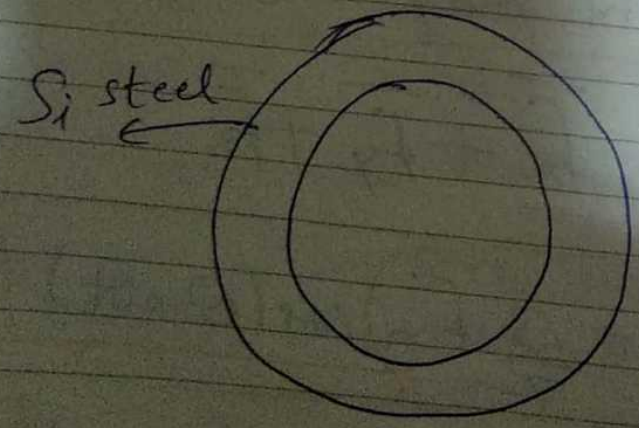
$\Rightarrow$ Construction of Induction m/c :-

It consists of :-

(1) Stator (2) Rotor

Rotor → Squirrel cage rotor, Slip ring rotor.

(1) Stator of Induction motor :-

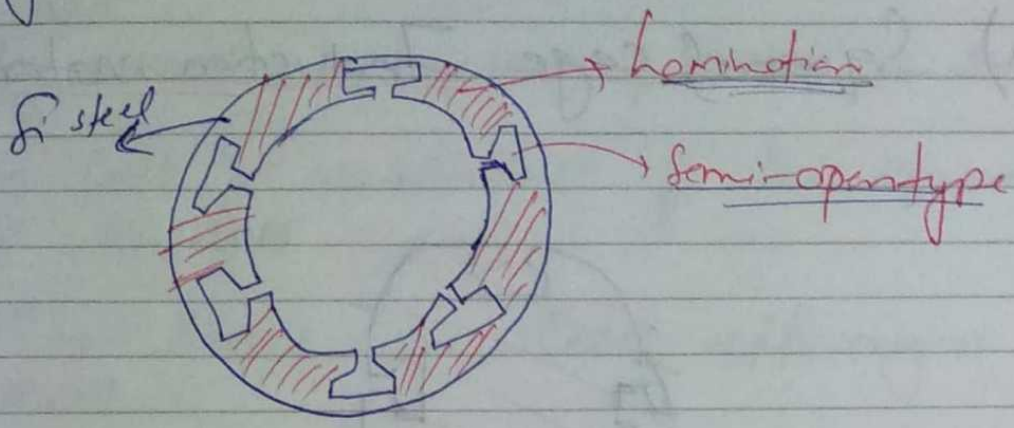


Stator is made up of Silicon(Si) steel. & the slots are of semi-open type of slots & it is laminated to reduce

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eddy current losses



→ The purpose of stator is a 3-ϕ winding connected in Δ is placed in the slots having 120° space angle displacement

Note 3-ϕ winding connected in Δ as harmonics will reduced & the phase current is also reduces so that working operation of ~~neutral~~ Induction m/c will be better. [As $I_{ph} = \frac{I_L}{\sqrt{3}}$, so size of conductor is also smaller]

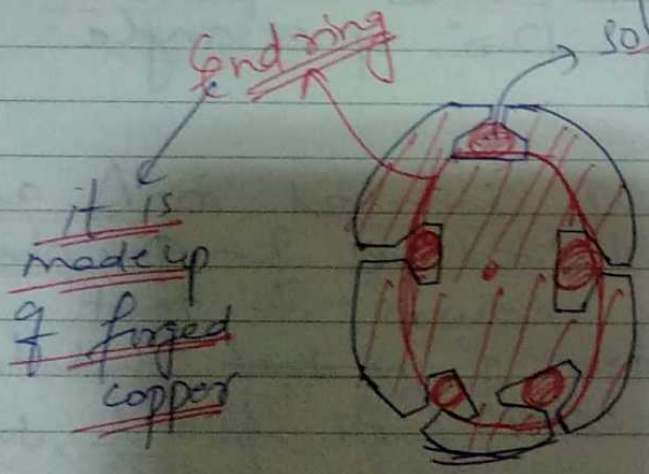
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② Rotor :-

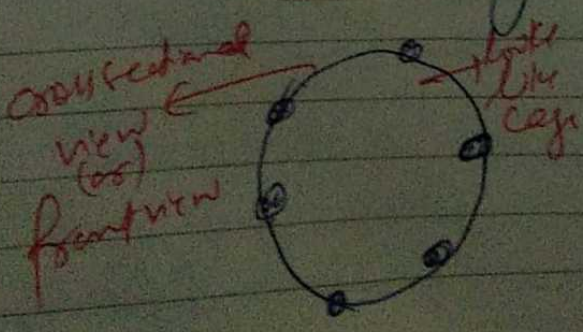
a) Squirrel cage Induction motor :-

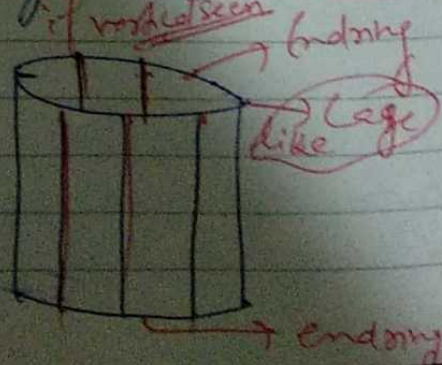




In this semi-open slots heavy (or) solid copper bars are placed (or) inserted & all the copper bars are short circuited with the help of end ring.

So, In order to operate Induction m/c, rotor winding always short circuited





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→ Advantages :

1. The winding procedure of rotor is simple.
2. The cost & maintenance of this squirrel cage is less.
3. The losses are less, efficiency is more.
4. Air Gap is less.
5. It is having good running performance.

⇒ Disadvantages :

(1) As rotor resistance is less, the starting Tst of motor is less.

{At ⓒ → solid copper bars so at no R}

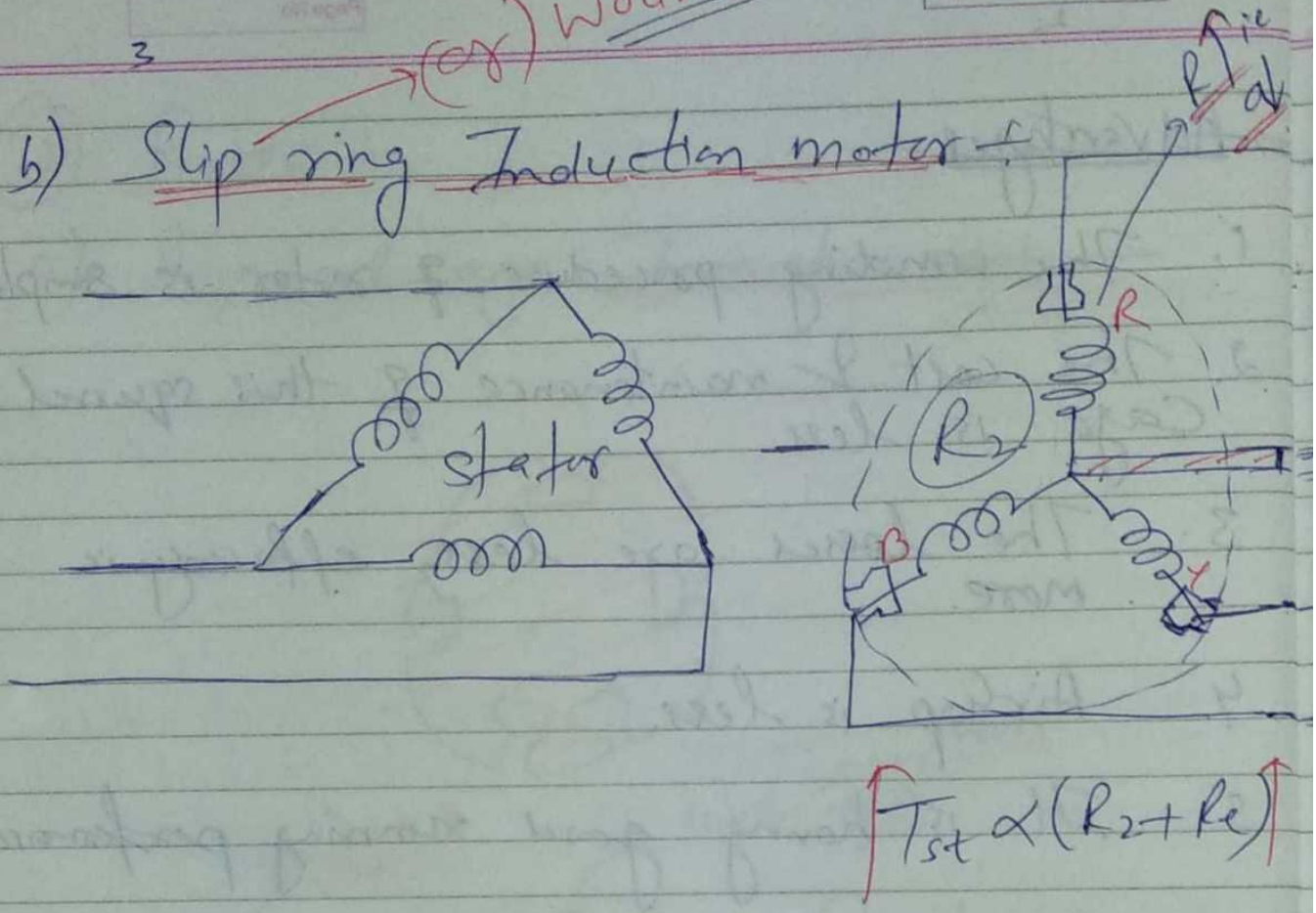
(2) The starting current Ist is very high which will damage the rotor wdg.

(3) Speed control methods are less.

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(or) Wound rotor

b) Slip ring Induction motor :-



$T_{st} \propto (R_2 + R_e)$

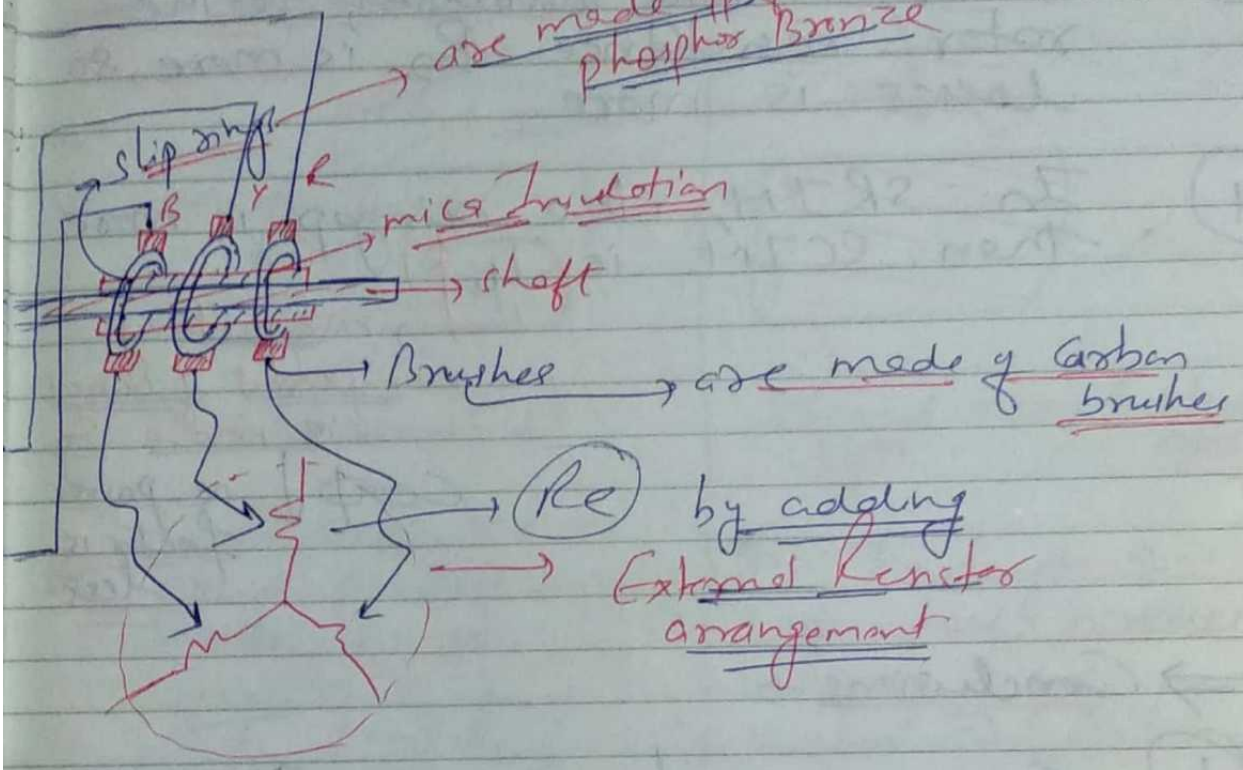
⇒ Advantages & SRIM :-

(1) As Rotor resistance is more, the T_{st} (starting Torque) is more & also by adding external resistance

Note for adding external resistor in series with each phase, the rotor winding should connect in star (Y).

if Δ connected → Not possible

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(2) By adding external resistor the starting current drawn by Induction motor can be limited.

(3) Speed Control Methods are simple (or) it is having more no. of methods.

⇒ Disadvantages :-

(1) The size & cost of Slip Ring Induction motor is more

(2) The Maintenance is more, i.e it is required to check regularly the brushes & slip ring arrangement.

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(3) Under running condition, as the rotor resistance R_2 is more, so losses is more

(4) In SRIM, the airgap is more than SCIM. ie (I_μ ↑), magnetising component of current is more So, $\cos\phi$ ↓ ie power factor is less

⇒ Conclusions :

(1) Compared to SRIM, the SCIM has better performance under running condition. Hence it is preferred mostly in domestic purpose.

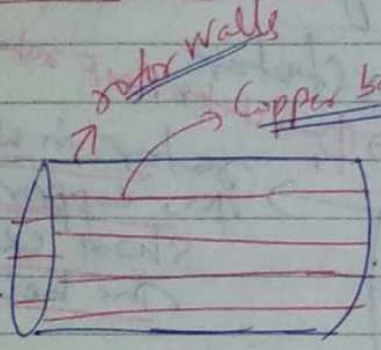
(2) SCIM has a good power factor compared to SRIM.

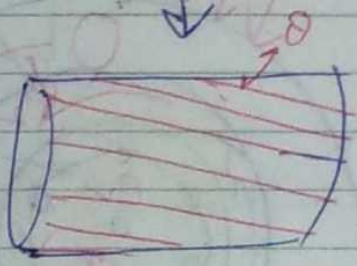
(3) The starting performance of SCIM can be improved by using Skewed Rotor.

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⇒ "Skewed Rotor" :-





↳ Normal SCIM

↳ here copper bars are placed parallel to rotor walls. hence

$$T_{st} \downarrow \rightarrow R_2 \downarrow$$

Skewed means twisted rotor

∴, the length of Copper bars increases as

$$L \uparrow \Rightarrow R \uparrow \propto L \uparrow$$

∴, as $R \uparrow \Rightarrow T_{st} \uparrow$

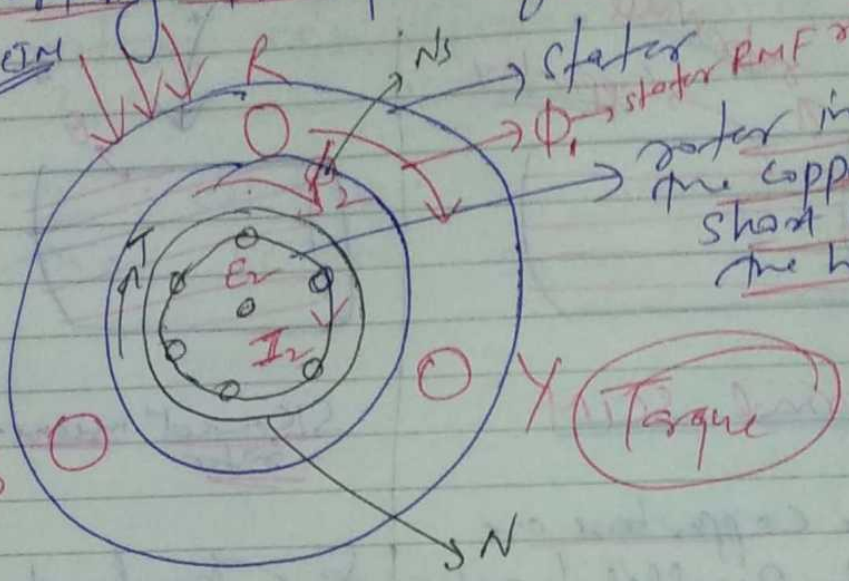
Note: Due to "skewed rotor" the starting torque improved (i.e $T_{st} \uparrow$) & also harmonics reduces due to which a smooth rotation of rotor is possible.

↳ means noiseless rotation

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⇒ Working Principle of Induction motor:



So when a 3-ϕ supply is given to stator, assume an mmf (rotating magnetic field) is flowing in stator i.e Φ_1 at synchronous speed Ns, due to which at starting mmf is rotating but rotor is stationary so due to rotating flux i.e Φ_1 an emf will induced in rotor i.e E_2 (as rotating flux cuts the conductor), & at the same time as rotor bars are short circuited so an current I_2 will flow in ~~rotor~~ copper bar which produce flux again Φ_2 which should act in same direction as Φ_1 due to Lenz's law. So due to these two fluxes, i.e stator mmf (flux) & rotor mmf (flux) a Torque will developed i.e torque will rotate in same direction, & torque will drive the rotor in same direction so rotor starts rotating

with Speed N. & less than N_s

ie $N < N_s$

→ So, Induction motor also called as "Asynchronous motor"

Note: when a 3-ϕ supply is given to stator, an RMF (ϕ_1) will produce rotating at N_s. Due to ϕ_1 an emf will induce in the rotor winding which causes a rotor flux ϕ_2. ∴ To satisfy Lenz's law, the rotor flux ϕ_2 will also rotates, in the same direc^n of ϕ_1 at N_s. During this a Torque will develop which draws the rotor in the same direc^n. But the rotor rotates at Speed, N always which is less than N_s. → ie $N < N_s$

∴ The relative speed b/w the rmf & rotor is $N_s - N$ → which is called slip speed

∴ slip speed of Induction motor = $N_s - N$

& in p.u slip, $s = \frac{\text{slip speed}}{N_s} = \frac{N_s - N}{N_s}$

per unit

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Name - Prashant Nigam

Sub - EM

↳ (Electrical Machines)

$$S = \frac{N_s - N}{N_s}$$

→ slip of Induction motor

(or)

$$S N_s = N_s - N$$

$$\Rightarrow \boxed{N = N_s (1 - S)} \text{ (rpm)}$$

→ Speed of motor in rpm

Note: So, Speed N depends upon two parameters

1) N_s → Synchronous Speed &

2) S → Slip

→ At starting

Let stator freq ≡ f

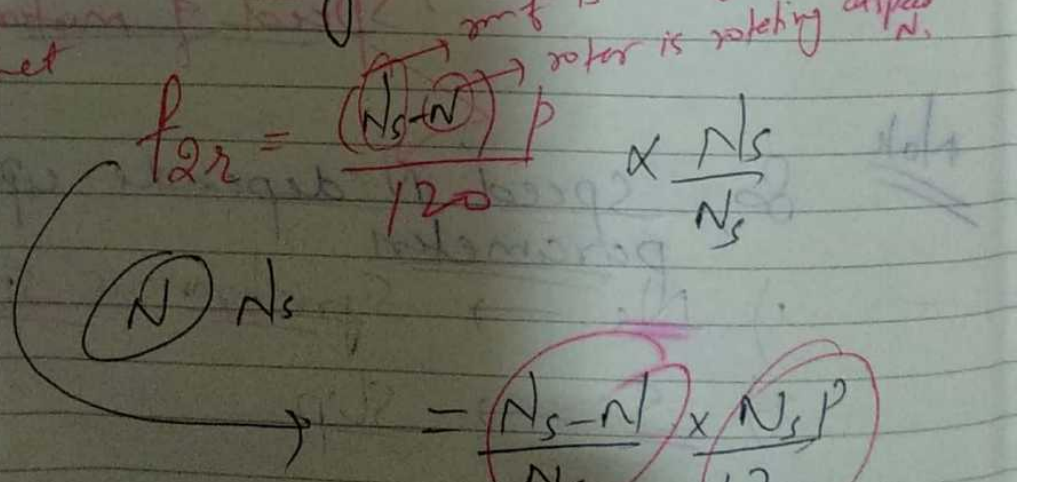
∴ Roter freq 'f_2' = ?

As $f_2 = \frac{(N_s)P}{120} = f$

$$\boxed{f_2 = f} \rightarrow \text{at starting condition}$$

But Under Running Condⁿ

Let



Note:

1. The speed of Rotor RMF w.r.t stator is N_s.

2. The speed of Rotor w.r.t stator is N.

3. The speed of Rotor w.r.t RMF is $(N_s - N)$.

4. The speed of stator mmf with respect to rotor mmf is $(N_s - N_s) = 0$.

⇒ Changes in Rotor parameters :-

→ Rotor parameters :-

1. R_2 → Rotor Resistance — it will remain same
2. X_2 → Rotor Reactance
3. $Z_2 = (R_2 + jX_2)$ → Rotor Impedance
4. E_2 → Rotor EMF
5. I_2 → Rotor Current

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At Starting

$$f_2 = f$$

$E_{2ph} = 4.44 B_m A_n f N_{2ph} K_{w2}$ → winding factor rotor

$Z_2 = R_2 + jX_2$

As $X_2 = 2\pi f L$

$|Z_2| = \sqrt{R_2^2 + X_2^2}$

$$\therefore I_2 = \frac{E_2}{\sqrt{R_2^2 + X_2^2}}$$

→ rotor current at standstill

Now Under Running Condⁿ :-

$$f_{2r} = sf$$

$E_{2r} = 4.44 B_m A_n (sf) N_{ph} K_{w2}$

(or)

$$E_{2r} = s[E_{2ph}]$$

$X_{2r} = 2\pi(sf) \cdot L = sX_2$

$Z_2 = \sqrt{R_2^2 + (sX_2)^2}$

$$\therefore I_{2r} = \frac{E_{2r}}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

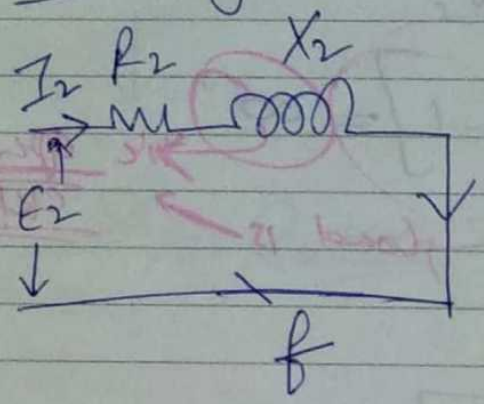
→ rotor current at running condⁿ

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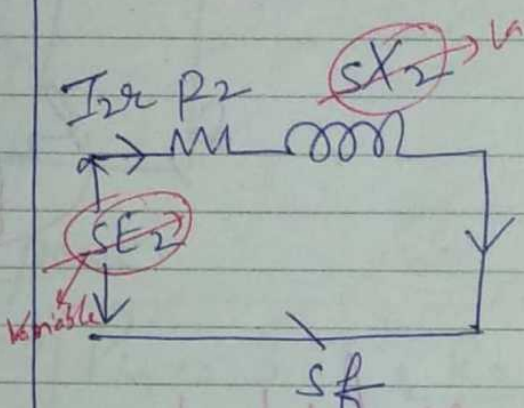
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⇒ Rotor Equivalent Ckt :-

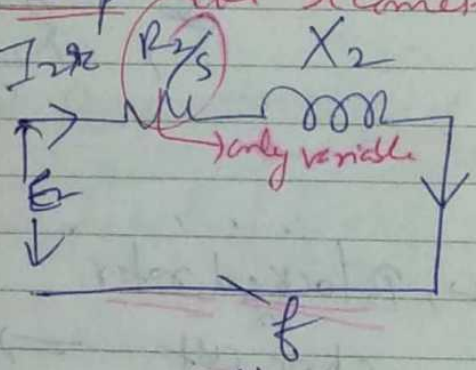
At starting

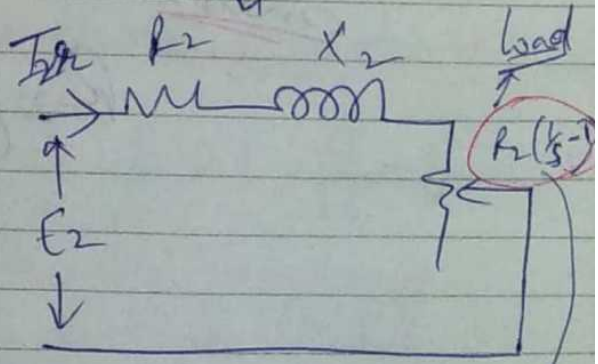


At Running



So, Dividing all the parameters of rotor with slip ckt becomes





So, $R_2\left[\frac{1}{s}-1\right]$ → mechanical load.

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&, At No-load $\Rightarrow$ $N = N_s$

have $S = \frac{N_s - N}{N_s} \Rightarrow S = 0$

&, $R_2\left[\frac{1}{S} - 1\right] = \infty$ $\Rightarrow$ i.e. open ckt

load ↑

Now, At full load $N \neq N_s$ $\Rightarrow$ then $S = 0.04, 0.06$

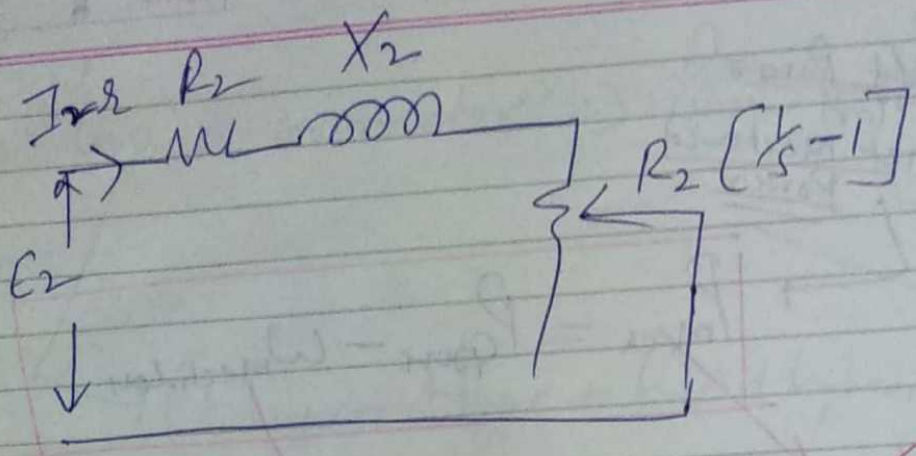
→ slip is low

&, $R_2\left(\frac{1}{S} - 1\right) \rightarrow$ high resistance

Now, Blocked rotor or short ckt condition $\Rightarrow N = 0 \rightarrow$ then $S = 1$

$R_2\left[\frac{1}{S} - 1\right] \Rightarrow 0 \rightarrow$ which indicates short ckt

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8, Rotor I/P power (for 3-ϕ)

$$P_{in} = 3 I_{2r}^2 \frac{R_2}{s}$$

Rotor Cu loss $\Rightarrow$ $P_{cu} = 3 I_{2r}^2 R_2$

2 The gross mechanical power developed

$$P_{gross} = 3 I_{2r}^2 R_2\left[\frac{1}{s}-1\right]$$

$\rightarrow$ Relation b/w Rotor Power Equations

$$P_{in} : P_{cu} : P_{gross} = \frac{1}{s} : 1 : \left(\frac{1}{s}-1\right)$$

$$P_{in} : P_{cu} : P_{gross} = 1 : s : (1-s)$$

8, if we want

$$\frac{P_{in}}{P_{cu}} = \frac{1}{s} \quad \left| \quad \frac{P_{in}}{P_{gross}} = \left(\frac{1}{1-s}\right) \right. \quad \left| \quad \frac{P_{cu}}{P_{gross}} = \frac{s}{1-s} \right.$$

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Sometimes ask shaft Power (or) net mechanical Power

$P_{mech} = P_{Gross} - W_{mech\,loss}$

P_{mech} → Shaft or net Mechanical Power

$W_{mech\,loss}$ → Mechanical loss means friction & windage loss

Now total losses in motor ⇒ $P_{input} - P_{net}$

= _____ watts

$(W_{stator\,Cu} + W_{rotor\,Cu} + W_i + W_{mech}) = (\quad)$

$W_i + W_i + W_i + \left(\frac{1}{3}\right)W_i =$ _____

$W_i =$ (_____) watts $= W_{stator\,Cu\,loss}$

Q A 3-ϕ, 40KW (indicates net mechanical (or) shaft power), SCIM has an efficiency of 90%. The stator & rotor copper loss each equals to iron loss & mechanical loss is $\frac{1}{3}^{rd}$ of Iron loss. Then find the stator Cu loss?

Sol: Given $\eta = 90\%$

& $\eta = \frac{P_{net}}{P_{ip}}$ (P_{ip} → motor I/P Power (or) stator I/P Power)

$90\% = 0.09 = \frac{40KW}{P_{ip}}$

Stator I/P Power $P_{ip} = \frac{40}{0.09} =$ _____

Q A 3-ϕ, 400V 50Hz Induction motor draws 50KW from supply. The rotor emf makes an oscillations of 100 osc/min If the stator losses are 2KW then find the rotor Cu loss _____?

Sol:



Rotor Cu loss = ?

Here Rotor I/P is $P_{ip} = 48KW$

Q Given $E_{2r} = \frac{100}{60} = \frac{5}{3}$

& rotor freq. = $sf = 5/3$

$\Rightarrow s[50] = 5/3$

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$\Rightarrow s = 1/30$

So, we have relation $\Rightarrow \boxed{\frac{P_{in}}{P_{cu}} = \frac{1}{s}}$

$\Rightarrow P_{cu} = P_{in} \times s$

$\Rightarrow (48KW) \times 1/30$

$\boxed{P_{cu} = 1.6KW}$ Ans

→ Rotor Cu loss

Q An 8-pole, 3-ϕ 50 Hz induction motor is operating at a speed of 700 rpm The frequency of the rotor current of the motor is _____ ?

Soln Given $P = 8$

$N = 700$ rpm

$\therefore N_s = \frac{120 \times 50}{8} = 750$ rpm

Now $s = \frac{N_s - N}{N_s} = \frac{750 - 700}{750} = 0.067$

$s = 0.067$

& $\boxed{f_2 = sf}$

→ $f_2 = 0.067 \times 50$

$\boxed{f_2 = 3.33 Hz}$

→ freq of rotor current

Q The slip of Induction motor normally doesn't depend on ___(d)___

a) Rotor speed b) Synchronous Speed

c) Shaft Torque d) Core loss Compon

Soln $s = \frac{N_s - N}{N_s}$ (or) $s = N_s(1 - s)$

Q A 3-ϕ, 50 Hz slip ring Induction motor has 6 poles on stator & 4 poles on rotor, then the possible speed of motor is _____ ?

a) 960 rpm b) 1440 rpm c) 1400 rpm

d) 0 rpm

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Note for different no. of poles on stator & rotor, the induction motor operation is not possible

Q The Power I/P to a 3-ϕ Induction motor is 70KW. The stator Cu loss & Iron loss are 2.5KW. find the mechanical power if the motor is running with a slip of 5%?

Sol: a) 67.5KW b) 64.125KW
c) 65KW d) 62.5KW

Given Stator I/P (P_{in}) = 70KW

Stator losses ⇒ 2.5KW

$S = 5\% = 0.05$

$P_{gross} = ?$

Ans We know that

$$\frac{P_{in}}{P_{gross}} = \frac{1}{1-S}$$

So Stator O/P ⇒ P_{in} − stator loss
→ 70 − 2.5 = 67.5KW

& $P_{gross} = P_{in}[1-S] \Rightarrow 67.5[1-0.05]$

$\boxed{P_{gross} = 64.125\,KW}$ Ans

Q The rotor Emf of a 3-ϕ, 6 pole 400V, 50Hz induction motor alternates at 3Hz. Then the Speed of motor is 940 rpm?

Sol: Given $P = 6$, $f = 50Hz$

$N_s = \frac{120f}{P} = \frac{120 \times 50}{6}$

$\boxed{N_s = 1000\ rpm}$

Rotor Emf → $\Rightarrow \boxed{3Hz = sf}$

$\Rightarrow S = \frac{3}{50} = 0.06$

$S = 0.06$

∴ Speed of motor

→ $N = N_s(1-S)$
$= 1000(1-0.06)$

$\boxed{N = 940\ rpm}$ Ans

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Q The power I/P to 415V, 50Hz, 6 pole 3-ϕ induction motor running at 975 rpm is 40kW. The stator losses are 1kW & mechanical losses are 2kW. Then efficiency of motor is 90%?

a) 92.5% b) 90% c) 91% d) 87%

Sol: Given

Stator I/P ⇒ 40kW

N = 975 rpm

η = ?

At $\eta = \frac{P_{net}}{P_{in}}$ → $P_{net} = P_{Gross} - W_{mech}$

∴ Stator O/P ⇒ Rotor I/P (P_{in}) ⇒ 40kW

∴ Rotor I/P (P_{in}) = 39kW

At $\frac{P_{in}}{P_{Gross}} = \frac{1}{1-S}$ ①

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⇒ $S = \frac{N_s - N}{N_s} = \frac{1000 - 975}{1000} = 0.025$

S = 0.025

& eq ① becomes

⇒ $P_{Gross} = P_{in}(1-S)$

$= 39[1 - 0.025]$

$P_{Gross} = 38.025$ kW

$P_{net} = P_{Gross} - W_{mech}$

$= 38.025 - 2$

$P_{net} = 36.025$ kW

Hence $\eta = \frac{36.025}{40} = 90\%$

η = 90% Ans

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⇒ Torque Eqn of Induction motor :-

Due to interaction b/w stator mmf ϕ_1 & rotor mmf ϕ_2

$\Rightarrow T \propto P_{gross}$

$\Rightarrow \boxed{T = \frac{P_{gross}}{\omega}} = \frac{3 I_{2r}^2 R_2 \left[\frac{1}{s} - 1\right]}{\frac{2\pi N}{60}}$

$= \frac{180}{2\pi N} 3 I_{2r}^2 R_2 \left[\frac{1-s}{s}\right]$

(or)

$T = \frac{180}{2\pi N} \left[\frac{s \cdot E_2}{\sqrt{R_2^2 + (sX_2)^2}}\right]^2 \frac{R_2}{s}\left[\frac{N_s - N}{N_s}\right]$

$T = \frac{180}{2\pi N} \left[\frac{s \cdot E_2}{R_2^2 + (sX_2)^2}\right] \cdot R_2 \left[\frac{N}{N_s}\right]$

$\therefore \boxed{T = \frac{180}{2\pi N_s} \left[\frac{s \cdot E_2^2 R_2}{R_2^2 + (sX_2)^2}\right] \frac{N_s}{0}}$

→ Generalised eqn of Torque

Note from eq ①

1. $\boxed{T \propto E_2^2 \propto G^2 \propto V_1^2}$

If the supply voltage V_1 is reduced by 10%, then

$T \propto (0.9V_1)^2 \propto 0.81 V_1^2$

→ T ↓ by 19%

2. $\boxed{T \propto s}$ ⇒ if slip is low (i.e low value 0.01, 0.02, 0.03 — 0.1)

$s = \frac{N_s - N}{N_s}$

if $N = N_s$, then $s = 0$

& if $N = 0$, then $s = 1$ → means just starting

& $\boxed{T \propto 1/s}$ ⇒ if slip is high (i.e s = 1) (like 0.9, 0.8, 0.7 0.6)

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⇒ Maximum torque = $(T_{max})_t$

$$T = \frac{180}{2\pi N_s}\left[\frac{S E_2^2 R_2}{R_2^2 + (SX_2)^2}\right]$$

→ also called Pull out torque (or) Breakdown torque (or) stalling torque.

→ It is defined as the maximum torque given by the induction motor at the edge it becomes unstable i.e pulling out the induction motor from one region to other region.

⇒ Condition for maximum torque:

$$\frac{dT}{ds} = 0$$

So, we get

$$SX_2 = R_2$$

hence

$$S_m = \frac{R_2}{X_2}$$

→ maximum slip corresponds to max. torque

∴ Maximum Torque is

i.e $T_{max} = \frac{180}{2\pi N_s}\left[\frac{S_m E_2^2 R_2}{R_2^2 + (S_m X_2)^2}\right]$

$$T_{max} \Rightarrow \frac{180}{2\pi N_s} \cdot \frac{R_2/X_2 \cdot E_2^2 R_2}{R_2^2 + \left[\frac{R_2}{X_2} \cdot X_2\right]^2}$$

$$= \frac{180}{2\pi N_s} \cdot \frac{E_2^2 R_2^2}{2 R_2^2 X_2}$$

$$T_{max} = \frac{180}{2\pi N_s} \cdot \frac{E_2^2}{2X_2}$$

→ max. torque produced in induction motor

Note

1. $T_{max} \propto E_2^2 \propto G^2 \propto V_1^2$

2. $T_{max} \propto \frac{1}{X_2}$ → rotor reactance depends on frequency

3. $(T_{max})_{SCIM} > (T_{max})_{SRIM}$

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⇒ Starting torque of Induction motor:

As at starting ↓ $N=0$ & $S=1$

$$\therefore T_{st} = \frac{180}{2\pi N_s}\left[\frac{1 \cdot E_2^2 \cdot R_2}{R_2^2 + 1 \cdot X_2^2}\right]$$

$$T_{st} = \frac{180}{2\pi N_s}\left[\frac{E_2^2 R_2}{R_2^2 + X_2^2}\right]$$

$R_2^2 \ll X_2^2$

Hence $T_{st} \uparrow \propto R_2 \uparrow$

→ only possible in Slip Ring Induction motor i.e. $T_{st} \uparrow \propto (R_2 + R_e)\uparrow$

Note

1. $T_{st} \propto E_2^2 \propto E_1^2 \propto V_1^2$
2. $T_{st} \propto \left(\frac{1}{R_2^2 + X_2^2}\right)$ (or) $T_{st} \propto \frac{1}{X_2^2}$

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⇒ Condition for maximum Starting torque:

At for max. torque $S_m = \frac{R_2}{X_2}$ — (1)

& for starting max. Torque $S=1$

& eq (1) become

$\Rightarrow 1 = \frac{R_2}{X_2} \Rightarrow$ i.e. $R_2 = X_2$

Note

1. $T_{st} = \frac{180}{2\pi N_s} \cdot \frac{S_{st} E_2^2 R_2}{R_2^2 + (S_{st} X_2)^2}$
2. $T_{max} = \frac{180}{2\pi N_s} \cdot \frac{E_2^2}{2X_2}$
3. $T_{st} = \frac{180}{2\pi N_s} \cdot \frac{E_2^2 R_2}{R_2^2 + X_2^2}$

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$$\frac{T_{ff}}{T_{max}} = \frac{2 \cdot S_m S_{ff}}{S_m^2 + S_{ff}^2}$$

→ decide the stability of Induction motor

$$\frac{T_{st}}{T_{max}} = \frac{2 S_m}{1 + S_m^2}$$

$S_m = \frac{R_2}{X_2}$

→ max. torque is taken as reference

Q) If 3-ϕ Induction motor has a full load slip of 80% of S_m, then, the T_{ff} in terms of T_{max} is _____ ?

Sol^n Given $S_{ff} = 80\%$ of S_m

& we know

$$\frac{T_{ff}}{T_{max}} = \frac{2 S_m \cdot S_{ff}}{S_m^2 + S_{ff}^2}$$

(or)

$$\Rightarrow \frac{T_{ff}}{T_{max}} = \frac{2}{\frac{S_m}{S_{ff}} + \frac{S_{ff}}{S_m}}$$

$$= \frac{2}{\frac{1}{0.8} + \frac{0.8}{1}} = 0.98 \text{ (or) } 98\%$$ Ans

Q) In a 3-ϕ Induction motor, the T_{st} developed is 120% & T_{max} developed is 200% w.r.t rated torque. Then slip corresponds to maximum torque is _____ ?

a) 0.33 b) 3 c) 3.25 d) 0.25

Sol^n we have to find S_m = ?

Given $T_{st} = 120\% \Rightarrow 1.2T$ & let rated torque

& $T_{max} = 200\% = 2T$

By we know

$$\frac{T_{st}}{T_{max}} = \frac{2 S_m}{1 + S_m^2}$$

$$\Rightarrow \frac{1.2T}{2T} = \frac{2 S_m}{1 + S_m^2}$$

$$\Rightarrow 1.2 S_m^2 + 1.2 = 4 S_m$$

$$\Rightarrow 1.2 S_m^2 + 1.2 - 4 S_m = 0$$

& after Solving we get

$S_m = 0.33$ & 3

& $S_m = 0.33$ Ans

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Q. A 3-ϕ, 50 Hz, 6 pole induction motor has a rotor resistance of 0.1Ω & reactance of 0.92Ω. Neglect the voltage drop in stator & assume that the rotor resistance is constant. Given $S_{fl} = 3\%$, the ratio of max Torque to full load Torque is _____ ?

Soln a) 1.587 b) 1.712 c) 1.94 d) 2.134

Sol: W.K.T

$$\frac{T_{fl}}{T_{max}} = \frac{2 S_m S_{fl}}{S_m^2 + S_{fl}^2}$$

& We have to find $\frac{T_{max}}{T_{fl}} = ?$

$$\frac{T_{max}}{T_{fl}} = \frac{S_m^2 + S_{fl}^2}{2 S_m S_{fl}} \quad \text{---(1)}$$

Given $R_2 = 0.1\Omega$ & $X_2 = 0.92\Omega$

& $S_m = \frac{R_2}{X_2} = \frac{0.1}{0.92}$

$$S_m = 0.108$$

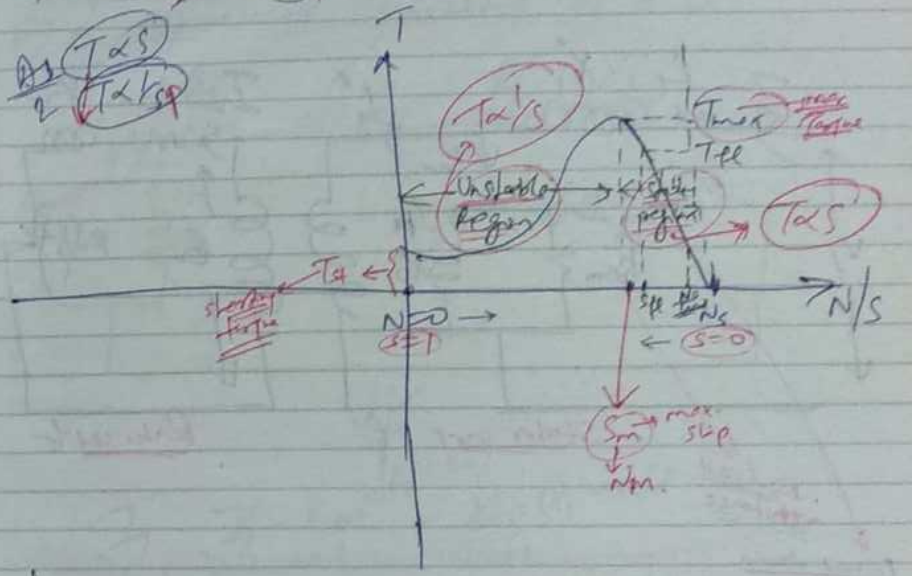
$\therefore$ eq (1) becomes $\frac{T_{max}}{T_{fl}} = \frac{(0.108)^2 + (0.03)^2}{2 \times 0.108 \times 0.03}$

$= 1.938 \simeq 1.94$ Ans

⇒ Torque - slip characteristics - is a fun^n of speed

$T \propto S$

$T \propto V_{sp}$

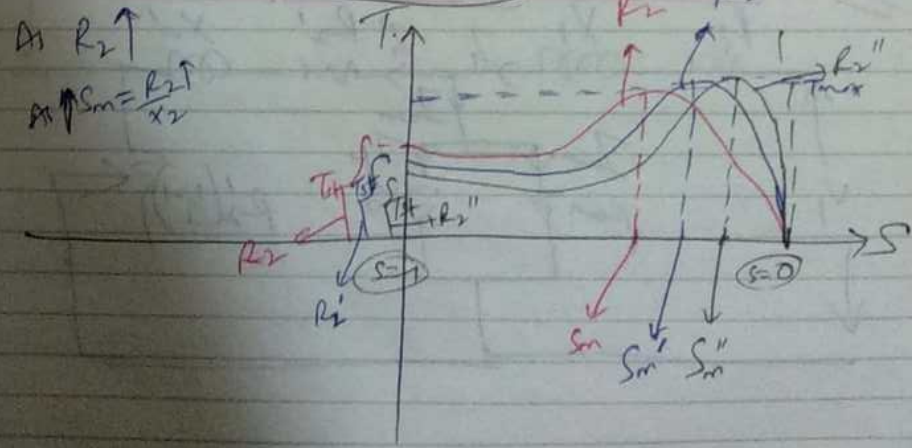


How Torque - slip characteristic for different rotor resistances -

Let $R_2 > R_2' > R_2''$

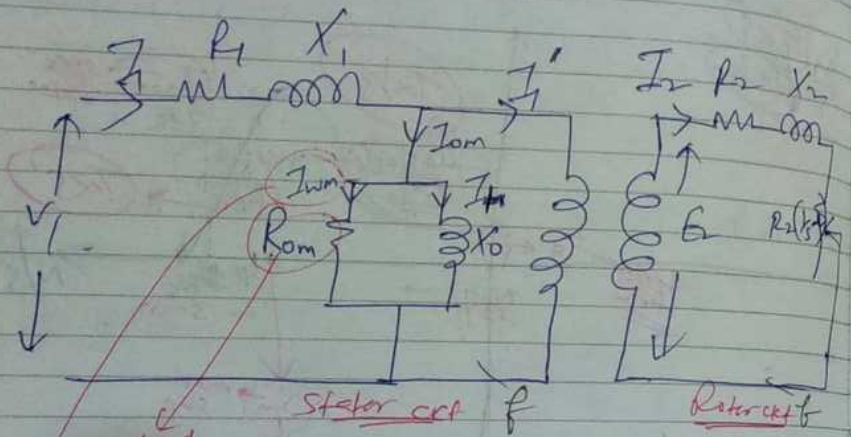
As $R_2 \uparrow$

As $S_m = \frac{R_2 \uparrow}{X_2}$



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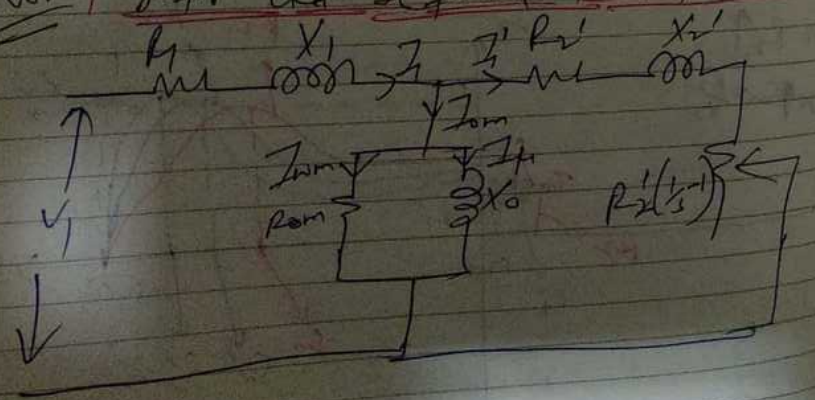
⇒ Equivalent circuit of Induction motor :-



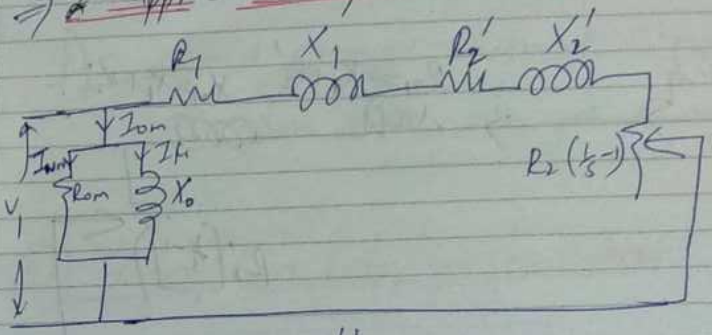
No load resistance

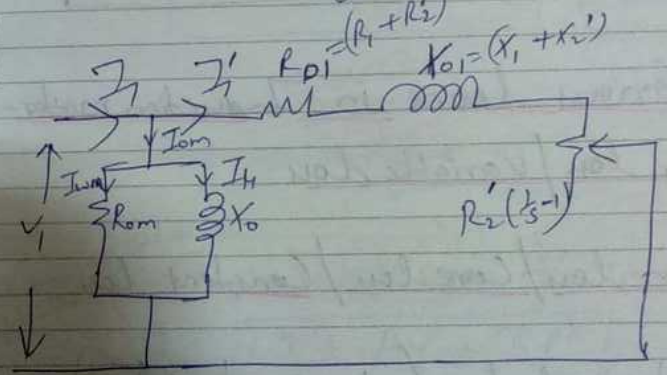
Current drawn by Induction motor in order to meet Iron loss & mechanical loss.

Now, rotor ckt referred to stator side



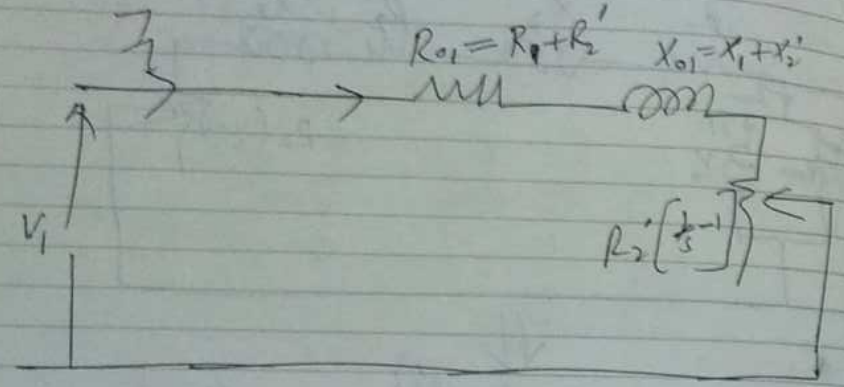
⇒ 2nd approximate Equivalent ckt :- i.e





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⇒ 2nd approximate Equivalent circuit :-



⇒ Various losses in Induction motor :-

1. Cu loss / Variable loss
2. Iron loss / Core loss / Constant loss
3. Mechanical loss (also constant loss)

① Cu loss / Variable loss :-

$I_1^2 R_1 + I_2^2 R_2$

∴ Total Cu loss $= I_1^2 R_1 + I_2^2 R_2$
$= I_1^2 [R_{01}]$ or $I_2^2 [R_{02}]$

∴ [Cu loss $\propto I^2 \propto$ load]

As load change, current change & Cu loss also get change

② Iron / Core loss / Constant loss

$W_i = W_h + W_e$

W_h → due to power & small circulating current in core

due to power & alternating retarded flux

W_e → always present in core

Case (I) When B_m = constant $\Rightarrow \frac{V_1}{f}$ = constant

∴ $W_i = Af + Bf^2$

Case II When $B_m \neq$ constant $\Rightarrow \frac{V_1}{f} \neq$ constant

∴ $W_i = AV_1^{1.6} f^{-0.6} + B \cdot V_1^2$

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Stator → f

& Running cond^n ⇒ stator → f

but rotor → (sf) very low → f_2

Due to this, the iron loss in induction motor is due to stator only. Since a rotor freq is low (negligible), Thus the iron loss also negligible.

③ Mechanical loss :

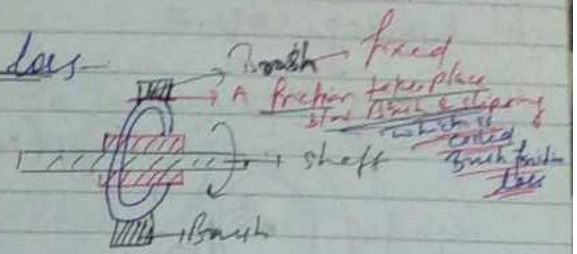
(a) friction losses

(b) Windage losses

due to → Brush friction loss (present in SRIM)

& → Bearing friction loss (present in both i.e SRIM & SCIM)

① Brush friction loss

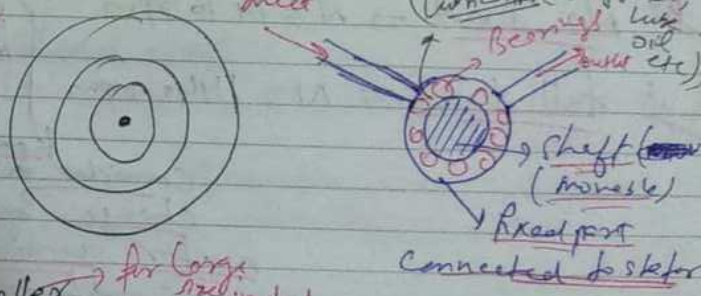


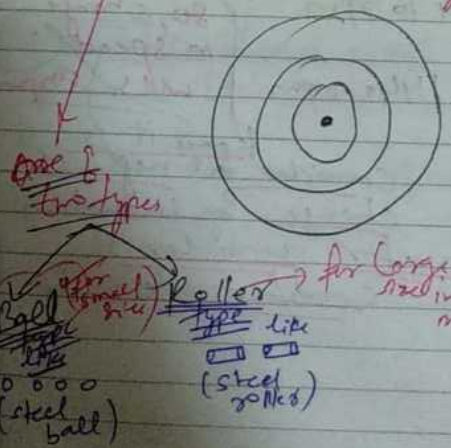
It is the friction takes place b/w the stationary brushes & rotating slipring

So, it can be reduced by using the Carbon Brushes → self lubricant material

② Bearing friction loss : → reduce this loss we add (lubricants)

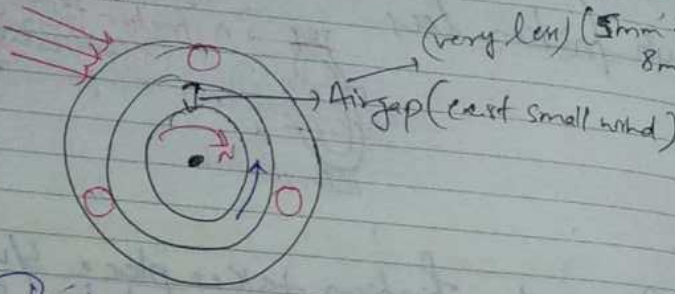
present in both induction motor





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b) Windage loss :- It is negligible loss in induction motor



During the rotation of rotor, the wind present in the airgap will oppose the direction of rotation which is called "Windage loss".

Note :- Friction & Windage loss depends upon speed of Rotor.

Let
At No load ⇒ N = 1490 rpm
& at full load ⇒ N = 1440 rpm

So, change in speed will be 50 rpm. Hence it is consider as negligible change, So induction motor consider as constant m/c motor.

Note :- The change in speed of induction motor from no-load to full load is negligible. Hence it is treated as constant speed motor.

∴ The mechanical loss which depends on speed is also treated as a constant loss.

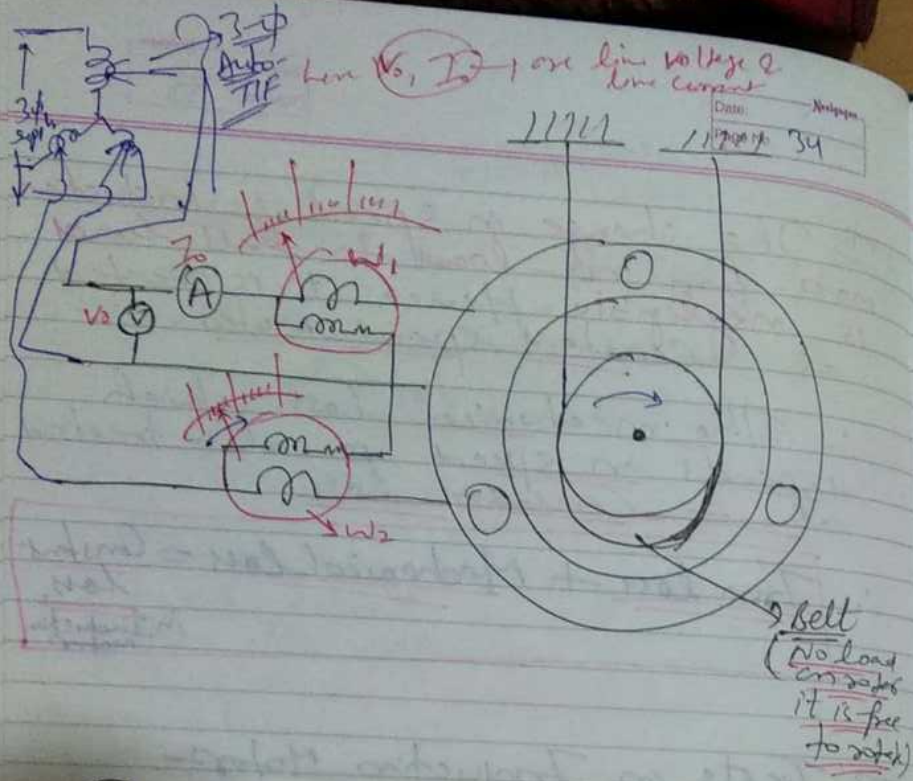
∴ Iron loss + Mechanical loss = Constant loss in Induction motor.

⇒ Tests on Induction Motor :-

1. No load test
2. Blocked Rotor test

(1) No load test :-
(It's purpose is to find Constant loss & to find no-load branch parameter i.e [Rom & Xo])

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Procedure

1. Connect the circuit as shown
2. observe the Induction motor should be under no-load
3. Apply rated voltage & rated frequency using 3-φ variac
4. Take the readings of Wattmeter (W_1, W_2) & Ammeter (A)

Total power consumed under no load

V_o	I_o	W_1 +ve	W_2 -ve	W_o $W_1 - W_2$

Note

Ckt No load pf ie $\cos\phi$ of induction motor under no load is 0.15 to 0.20

& when $\cos\phi < 0.5$ then W_1 = +ve deflⁿ (in forward direⁿ), W_2 = -ve deflⁿ (means backward direⁿ)

then we cannot read any value & interchange any two terminal of current coil (or) voltmeter coil then wattmeter rotation in forward direⁿ

a) to find ~~Constant~~ loss =

$W_o = (W_1 - W_2)$ = Constant loss + a small $3I_o^2 R_1$ loss (for I-φ) → Cu loss

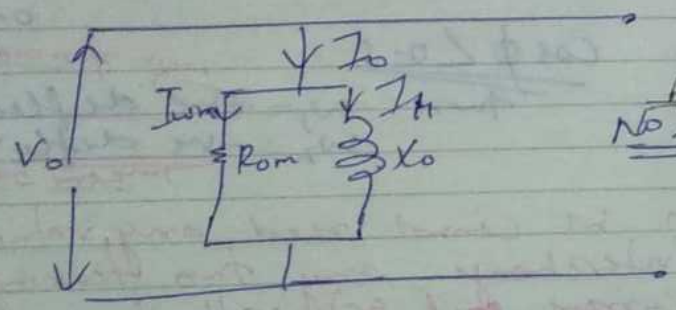
∴ Constant loss = W_o

(Since by neglecting small Cu loss)

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5) To find No load branch parameters [R_{om} & X_o]

f, Consider I^{ct} approximate ckt



$$\Rightarrow I_{wm} = \frac{V_{oph}}{R_{om}} \Rightarrow \boxed{R_{om} = \frac{V_{oph}}{I_{wm}}}$$

$$\text{Ig } \boxed{X_o = \frac{V_{oph}}{I_\mu}}$$

$$\text{2} \quad I_{wm} = I_{om_{ph}} \cos\phi_o \qquad W_o = 3 V_{oph} I_{oph} \cos\phi_o$$

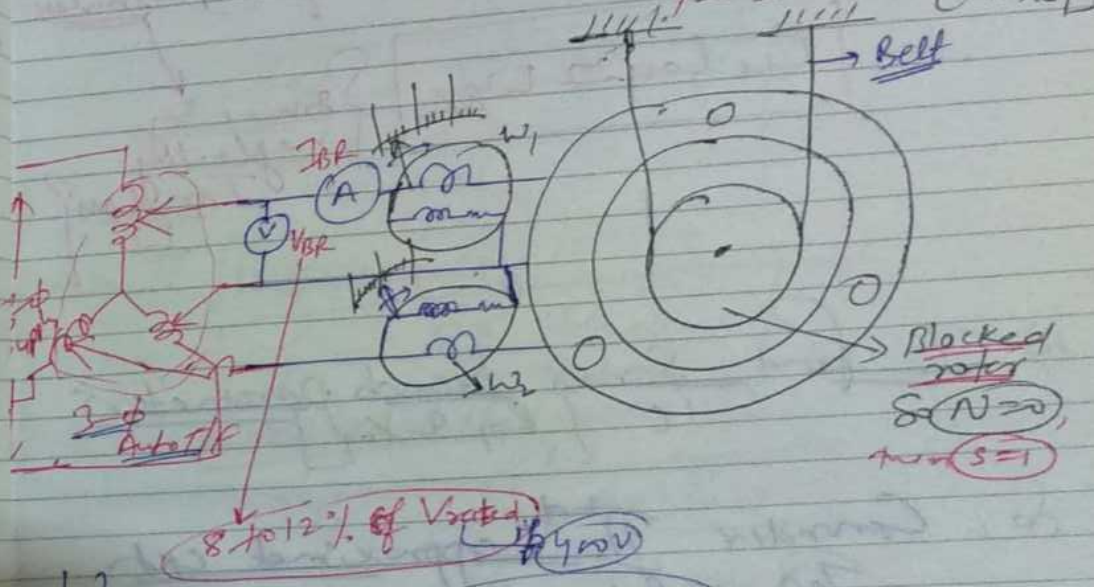
$$I_\mu = I_{om_{ph}} \sin\phi_o \qquad \boxed{\cos\phi_o = \frac{W_o}{3 V_{oph} I_{oph}}}$$

A I_{om} → line current. So convert it to phase current ie $\frac{I_{om}}{\sqrt{3}}$

(2) Blocked Rotor test :-

Should conduct at Rated Current

Purpose is to find the Cu loss (or) Variable loss. & to find the series branch parameters [ie R_{o1} & X_{o1}]



8 to 12 % of V_{rated} if 400V

Under this test pf $\cos\phi_{BR} = 0.6$ lag (0.5)

Both wattmeter readings (the) should be deflected

V_{BR}	I_{BR}	W_1	W_2	W_{BR} = $W_1 + W_2$

→ Total power consumed under blocked rotor

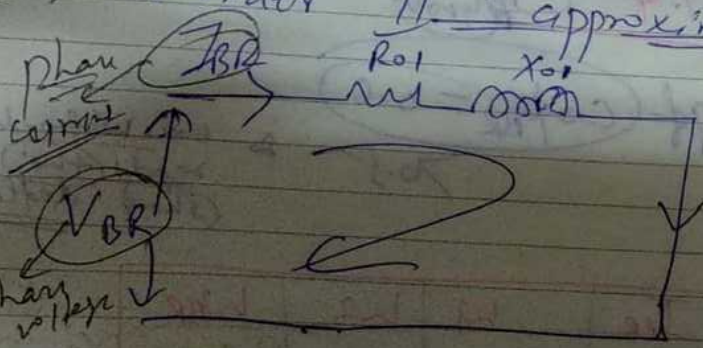
a) To find 'Cu loss :

$\Rightarrow W_{BR} = (W_1 + W_2) =$ 'Cu loss + a small amount of W_i (iron loss)

$\Rightarrow$ ['Cu loss = W_{BR}] { since by neglecting Iron loss W_i }

b) To find Series branch parameters ie [R_{01} & X_{01}]

&, Consider II nd approximate ckt.



So, $V_{BR} = I_{BR}[R_{01} + jX_{01}]$

& [$Z_{01} = \frac{V_{BR}}{I_{BR}}$]

&, [$W_{BR} = 3 I_{BR}^2 R_{01}$]

& [$R_{01} = \frac{W_{BR}}{3 I_{BR}^2}$]

$\therefore$ [$X_{01} = \sqrt{(Z_{01})^2 - (R_{01})^2}$]

$\Rightarrow$ Efficiency (η) :

[$\eta = \frac{\text{O/p Power}}{\text{I/p Power}}$]

(or)

[$\eta = \frac{P_{shaft} \text{ (or) } P_{net}}{P_{shaft} + \text{Losses}}$]

(or)

[$\eta_{Motor} = \frac{P_{shaft}}{P_{shaft} + W_i + W_{cu}}$]

W_i : Calculated from No load test

W_{cu} : Calculated from blocked rotor test

Includes both iron & mechanical loss

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Note: $\eta_{IM} < \eta_{TIP}$ → Transformer

Induction motor

⇒ Speed Control methods of Induction Motor :-

A) $N = N_s(1-s)$ → slip of induction motor

Synchronous Speed

Speed of motor

⇒ Purpose of controlling Speed of Induction Motor

a) To operate the loads smoothly

b) To save the amount of power

A) ~~$P = T\omega$~~

$\Rightarrow P = T\omega \Rightarrow T \times \frac{2\pi N}{60}$

So, $P \propto N$

Speed control Methods are :-

1) slip control methods
2) Synchronous speed Control methods

a) Voltage control method
b) Rotor resistance Control
c) Emf injecting Control

$N_s = \frac{120f}{P}$

Pole Changing Technique

Slip control method

① Slip control methods :-

a) Voltage control method ⇒

A) $T_L \propto \frac{s \cdot E_2^2}{R_2} \propto \frac{s \cdot V_1^2}{R_2}$

T_L always constant

$N = N_s(1-s)$

So, only $N < N_{rated}$ → possible in this method

Note: In any motor if voltage increases then it may possible insulation failure or damage so in order to increase the voltage in alternator or synchronous motor, level

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∴ Insulation must be proper.

∴ when the voltage reduces from the constant dip of induction motor should increase in order to maintain the load torque constant, then the speed of motor reduces.

Note: for above rated speeds, it is required to increase the supply voltage, but there is a chance of insulation failure, hence it is not preferred.

(b) Rotor resistance Control method:-

→ It is suitable for only "slip ring Induction motor"

Since in squirrel cage Induction motor it is not possible to vary rotor resistance.

$$T_2 \propto \frac{sE_2^2}{[R_2 + R_e]}$$

Constant → we are adding resistance slip also increases

$$\therefore N = N_s(1-s)$$

∴ only $N < N_{rated}$ below rated speed is possible

when an external resistance is connected in series with the rotor resistance R_2, then resultant resistance increases, but to maintain the load torque constant, the corresponding slip of induction motor should increase, Hence the speed of motor reduces.

Drawback of Rotor Resistance Control Method

(1) It is not suitable for above rated speed

(2) It is not suitable for SCIM

(3) Due to addition of resistance, the Copper loss of rotor increases which reduces efficiency of motor.

(c) Emf Injecting control:-

$$T_2 \propto \frac{sE_2^2}{R_2}$$

∴ the Emf into rotor can be inserted in two ways

1) In phase injecting method.

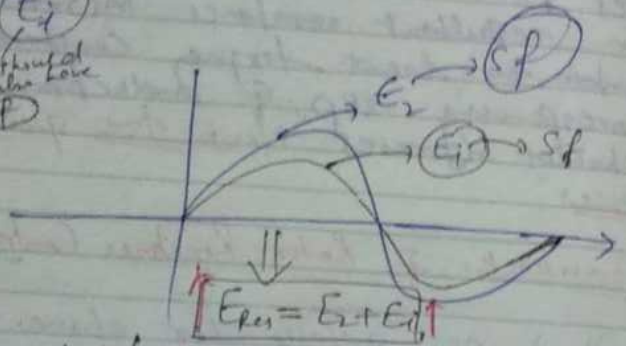
2) Out of phase injecting method.

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mean (0° displacement)

1) In phase Injecting method :-

Suppose this emf is added to rotor ckt



So, to add/inject an external emf (E_i) to the rotor emf E_2, its freq. should be same as that of slip frequency (Sf) otherwise it is not possible to add two signals.

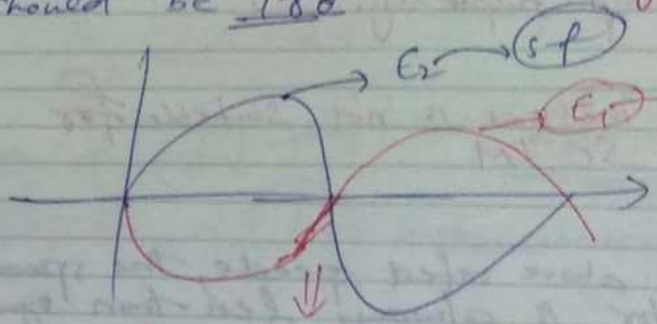
So, $T_d \propto \frac{S(E_2 + E_i)}{R_2}$

Hence $N = N_s(1 - S)$

So $N > N_{rated}$ above rated speed is possible

2) Out of phase Injecting Control :-

In this method, the phase displacement b/w rotor emf 'E_2' & Injecting emf E_i should be 180°



So, $E_{Ri} = E_2 - E_i$

$T_d \propto \frac{S(E_2 - E_i)^2}{R_2}$

Torque not Constant

$N = N_s(1 - S)$

$N < N_{rated}$

So here speed possible is below rated.

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As Drawback, (~~EMF injecting method~~) are :-

i) It is difficult to produce a voltage signal having frequency equal to slip frequency.

ii) This method is not suitable for SCIM

Note In above rated speeds, the speed of Motor is always less than synchronous speed only.

→ Rated speed is the speed of motor at rated current.

(2) Synchronous Speed Control methods

a) Frequency Control method:

As $N_s = \frac{120f}{P}$

& $N = N_s(1-s)$

($N \approx N_{rated}$)

By $N_s = \frac{120f}{P}$

& $N = N_s(1-s)$

&, ($N < N_{rated}$)

Suppose if ($f \downarrow$) if flux reduced so core get saturated.

& $B_m \propto \frac{V \downarrow}{f \downarrow}$ → & this ratio remain constant

↳ it get saturated

→ Above rated speed :-

$N_s = \frac{120f}{P}$, $N = N_s(1-s)$

($N > N_{rated}$)

So, when frequency of supply increases, the N_s & the motor will also increased. Due to which the speed of motor also increase. During this speed control, the $B_m \downarrow$ (reduces)

$B_m \propto \frac{V}{f}$

here, this method is called as "Pure frequency controlled method".

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⇒ Below Rated Speed :-

$$N_s = \frac{120f}{P}$$

$$N = N_s(1-s)$$

$N < N_{rated}$

$\phi_m \propto \frac{V_1}{f}$ ↓

constant ↓

∴ when the Ref. of supply reduces the speed of motor will also reduce simultaneously but during this the flux density of core increases, hence to avoid this saturation reduce the voltage also such that $\left(\frac{V_1}{f}\right)$ = Constant

∴ This method is called Variable voltage & variable frequency control method.

$$T_L \propto (N_s - N)\left(\frac{V_1}{f}\right)^2$$

T_L = constant

ii) If $\left(\frac{V_1}{f}\right)$ = constant therefore $(N_s - N)$ = constant

ii) If $\frac{V_1}{f} \neq$ constant, $(N_s - N)\left(\frac{V_1}{f}\right)^2 \neq$ Constant

Total product

Q A 3-ϕ, 400V, 50Hz SCIM driving a load at a speed of 960 rpm. If the voltage & frequency are reduced to 320V, 37.5 Hz, then the speed of motor in order to maintain the constant load torque is _____?

Sol^n Given 400V, 50Hz → $N_1 = 960$ rpm

$\frac{V_1}{f} = \frac{400}{50} = 8$ constant

& poles = 6

Now V = 320V, 37.5Hz → $\frac{320}{37.5} = 8 = \frac{V_2}{f_2}$ constant

$N_{s1} - N_1 = N_{s2} - N_2$

$\Rightarrow 1000 - 960 = \left[\frac{120 \times 37.5}{6}\right] - N_2$

& constant $\Rightarrow N_2 =$ _____ rpm Ans

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b) Pole changing method :

↳ Suitable only for SCIM. as here stator poles are dependent on rotor poles, rotor poles are dependent on stator poles.

$$N_s = \frac{120f}{P}$$

$$N = N_s(1-s)$$

If (P=4) → Now if Poles ↑ to 6

then N_s ↓ hence N ↓

So, when the poles of Induction motor increases so speed of motor N ↓

Ans When the poles of Induction motor decrease so speed of motor N ↑

Note Here, In this method the speeds obtain are the fixed speeds. So, it is not possible to get the smooth variation of speed.

Q This method suitable only to SCIM. Since the poles in SCIM are induced poles, but in case of SRIM the rotor poles are designed for fixed ones. Hence, with the changes in stator poles, the rotor poles will not change.

Note Mostly we prefer voltage control (or) frequency control methods.

Q A -3-ϕ, 6 pole 50Hz induction motor is operating with a rotor resistance of 0.2Ω/phase & running at 950 rpm. if it is required to operate at 800 rpm find the external resistance required to add in the rotor circuit is _____?

Sol^n Given No. of poles = 6
f = 50Hz
N = 950 rpm
R_2 = 0.2Ω/ph

N_2 = 800 rpm
R_{ext} = ?

Ans $T_d \propto \frac{S}{(R_2+R_{ext})}$ } in Rotor resistance control method

$N = N_s(1-S)$

So, $\frac{S}{R_2}$ = constant ①

$N_s = \frac{120f}{P} = \frac{120\times 50}{6} = 1000$ rpm

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$$S_1 = \frac{N_s - N_1}{N_s}$$

$$S_1 = \frac{1000 - 950}{1000} = \underline{0.05}$$

By 2 $S_2 = \frac{N_s - N_2}{N_s} = \frac{1000 - 800}{1000}$

$$= \frac{200}{1000} = \underline{0.2}$$

So from eq (1) we can write,

$$\Rightarrow \boxed{\frac{S_1}{R_2} = \frac{S_2}{R_2 + R_{ex}}}$$

$$\Rightarrow \frac{0.05}{0.2} = \frac{0.2}{0.2 + R_{ex}}$$

$$\Rightarrow \boxed{R_{ex} = 0.6\,\Omega/ph} \quad \underline{\text{Ans}}$$

Q The rotor emf of a 3-ϕ SRIM is 200V running at 950 rpm. If an emf of 20V is injected in out of phase with the rotor emf then the speed of motor to maintain the constant load torque is ____?

Sol^n Given $E_2 = 200\,V$

$N_1 = 950$ rpm

$E_i = 20\,V$ → out of phase (i.e. 180 displacement)

then $N_2 = ?$

So, During Emf injecting method

$T_L \propto S E_2^2$

So, $\boxed{S E_2^2 = \text{Constant}}$ → in order to get T_L = constant

So, $\boxed{S_1 E_2^2 = S_2 (E_2 - E_i)^2}$

Assume $N_s = 1000$ rpm (the nearest to speed of motor)

So, $S_1 = \frac{1000 - 950}{1000} = 0.05$

$0.05[20]^2 = S_2[20 - 2]^2$

$S_2 = 0.06$

So, $N_2 = N_s(1 - s)$

$= 1000(1 - 0.06)$

$\boxed{N_2 = 940\ \text{rpm}}$ Ans

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Q. In a 3φ Induction motor, if the supply voltage is reduced by 20% then the % decreased in torque is
a) 20% b) 40% c) 36% d) 10%

Sol:

$$T_d = \frac{180}{2\pi N_s}\left\{\frac{s E_2^2 R_2}{R_2^2 + (sX_2)^2}\right\}$$

$\therefore T \propto E_2^2 \propto E_1^2 \propto V_1^2$

If $V_1 \downarrow$ by 20% $\Rightarrow T \downarrow$

Suppose $T \propto 1.V_2^2$

i.e $T \propto (0.8V_1)^2$ (as voltage reduced by 20%)

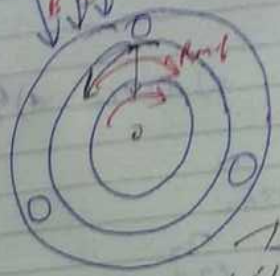
$T \propto 0.64 V_1^2$

$\therefore$ 0.36 ↓ (now remaining)

i.e T is reduced by 36%.

Q. In a 3-φ IM is operating at slip s. If 2 supply leads are interchanged then its slip at that instant is
a) s b) 1−s c) 2−s d) 1+s

Sol:



$s' = \frac{N_s - N}{N_s} = 1 - \frac{N}{N_s}$ i.e. $1 - s$

So, when two leads are interchange then the direction of RMF will reverse instantly, but not the direction of rotor

$\therefore$ Relative speed is $= N_s + N$

i.e backward slip $= (2 - s)$

(s_b)

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Q A 4-pole induction motor supplied by a slightly unbalanced 3-ϕ 50Hz supply is running at 1440 rpm. The freq. of induced -ve sequence current in the rotor is _____?

a) 100Hz b) 98Hz c) 52Hz d) 48Hz

Sol: Given pole = 4, f = 50Hz

$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500$ rpm

$s = \frac{N_s - N}{N_s} = \frac{1500 - 1440}{1500} = \frac{60}{1500} = \frac{1}{25} = 0.04$

Note: whenever there is an unbalanced 3-ϕ supply then the backward (or) -ve sequence currents are possible.

Forward slip

W.K.T $S_f = s \Rightarrow f_{2r} = s \cdot f$

$S_b = 2 - s \Rightarrow f_{2r} = (2-s)f$

Backward slip

$f_{2r} = (2-s) \cdot f$

$= (2 - 0.04) \times 50 = 98$ Hz

Q The blocked rotor current in 3-ϕ Y connected, 15KW, 4-pole, 230V, 50Hz IM at rated condition is 50A. Neglecting the losses & magnetizing current, the approximate blocked rotor line current drawn when the motor is connected to 236V, 57Hz supply is _____?

a) 58.5A b) 45A c) 45.70A d) 57.6A

Sol: Given $I_1 = 50$A

230V, 50Hz, P = 4

& power rating P = 15KW

$I_2 = ?$

236V, 57Hz

& W.K.T $I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}}$

Since blocked rotor & (s=1), &

$I_2 = \frac{E_2}{\sqrt{R_2^2 + X_2^2}}$

& losses neglected & ($R_2 = 0$)

&, $I_2 = \frac{E_2 \alpha}{X_2 \alpha} \Big|_f^V$

i.e $I_2 \propto \frac{E_2}{X_2} \propto \frac{V}{f}$

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So, $I \propto \frac{V}{f}$

Hence $\frac{I_2}{I_1} = \frac{V_2}{V_1} \times \frac{f_1}{f_2} \Rightarrow I_2 = \frac{236}{230} \times \frac{50}{57} \times 45$

$\Rightarrow I_2 = 45A$ Ans

Q A 3-ϕ, 440V, 50Hz IM is running at rated speed with a fixed load. If the supply voltage & frequency both reduced by Half (1/2). Then the max. torque developed in motor is _____?

Sol: a) reduced by 1/2

~~b)~~ remain same

c) Increased by 1/2

d) Doubles

Given $T_{max} = \frac{180}{2\pi N_s}\left\{\frac{E_2^2}{2X_2}\right\}$

$\Rightarrow T_{max} \propto \frac{V_1^2}{f^2} \propto \frac{(V_1)^2}{(f/2)^2} \propto \left(\frac{V_1}{f}\right)^2$

Hence T_{max} = constant (or) remain same Ans

Suppose By keeping (V = constant), if supply freq. is doubles Tmax is _____ Tmax ?

So, $T_{max} \propto \left(\frac{V_1}{f}\right)^2$

If freq. doubles

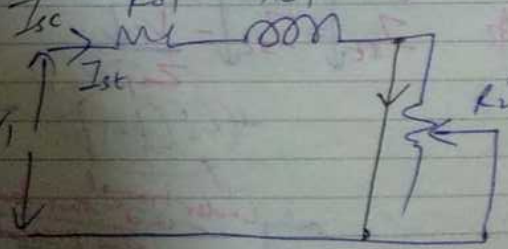
So $T_{max} \propto \frac{1}{4}\left(\frac{V_1^2}{f^2}\right)$

So T_{max} reduced by 1/4.

⇒ Starting methods of 3-ϕ Induction Motors

Note: Starting methods are required to limit the high starting current so that windings of motor may not get damaged.

Consider equivalent ckt of IM referred to stator.



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at the time of starting $N = 0$, $s = 1$

$s = \frac{N_s - N}{N_s} = \frac{0 - 1}{1}$

Replace $s = 1$, in $R_2' \left[\frac{1}{s} - 1\right]$

By applying KVL

$V_1 = I_{sc}[R_{01} + jX_{01}]$

$\Rightarrow V_1 = I_{sc} Z_{01}$

$I_{sc} = \frac{V_1}{Z_{01}}$ all are phasor values

$I_{sc} \rightarrow$ starting current

$\rightarrow$ 5 to 8 times I_{fl}

Q) because of this high current the winding will damage.

A) $I_{sc} = I_{sc} = \frac{V_{ph}}{Z_{eq} + Z_2'\left(\frac{1}{s} - 1\right)}$ it is fixed and low value

only current this term is varied after starting

only possibility to reduce the supply voltage to limit the starting current.

In order to limit the starting current, from the high starting currents, it is preferred to reduce the supply voltage at the time of starting.

Methods of starting

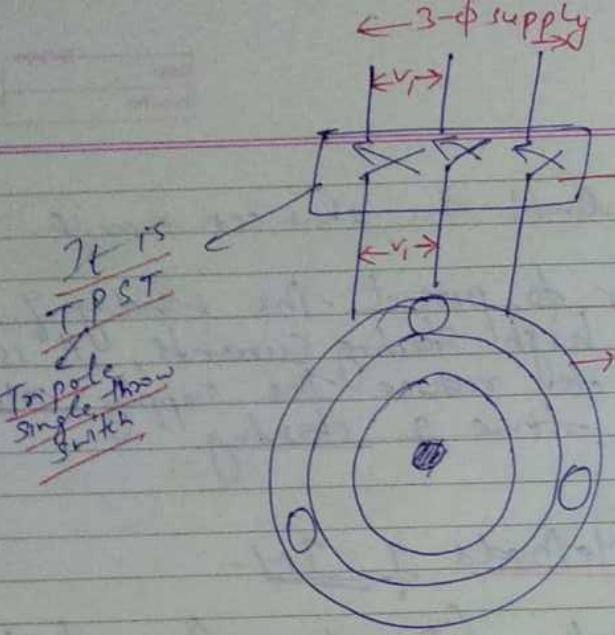
1) DOL starting (Direct on line starting)
2) Series reactor starting
3) Series resistor starting
4) Auto-T/F starting → performed in two steps
5) Y-Δ starting methods → preferred for small and medium purpose

This method is restricted to small rating motors.

① DOL starting method

$\left[\frac{I_{sc}}{I_{fl}}\right] \propto \left[\frac{I_{sc}}{I_{fl}}\right]^2 \cdot s_{fl}$

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It is TPST ie Triple pole Single throw switch

$$\therefore I_{st} = \frac{V_1}{Z_{01}} \rightarrow = I_{sc}$$

starting current

So, $\boxed{I_{st} = I_{sc}}$

Hence, In DOL starting method, once the starter is operated then the total supply voltage will appears across the stator of Induction motor, Hence the starting current drawn in this method is same as that of Short ckt current ie (I_{sc})

$$\therefore \boxed{\frac{T_{st}}{T_{fl}} = \left(\frac{I_{sc}}{I_{fl}}\right)^2 \cdot S_{fl}}$$

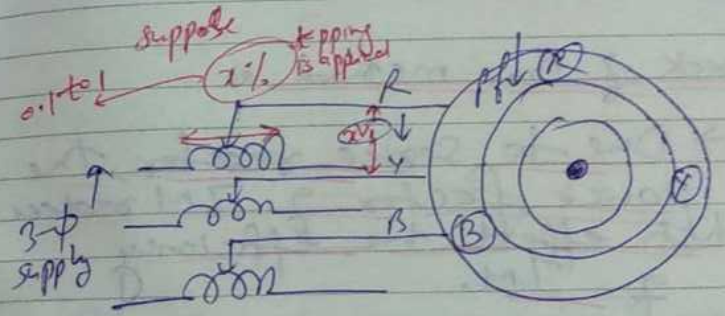
So, In this method there is no limit on short ckt current But it is considered as reference

Note

So, This method is taken as reference to compare with other methods

(2) Series Reactor starting method :-

→ Here we place three reactors in series with stator.



If x% tapping is applied on Series reactor then, the voltage across the stator appeared is also "xV1" which is reduced by fraction x. Hence the

$$\boxed{I_{st} = \frac{xV_1}{Z_{01}} \rightarrow I_{sc}}$$

$$\rightarrow (\because) \quad \boxed{I_{st} = x\,[I_{sc}]}$$

$\therefore$ Compared to DOL method the I_{st} drawn in this method is reduced by a fraction "x". i.e.

$$\frac{T_{st}}{T_{fl}} = \left[\frac{xI_{sc}}{I_{fl}}\right]^2 \cdot S_{fl}$$

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$$\frac{T_{st}}{T_{fl}} = x^2\left[\frac{I_{sc}}{I_{fl}}\right]^2 s.fl$$

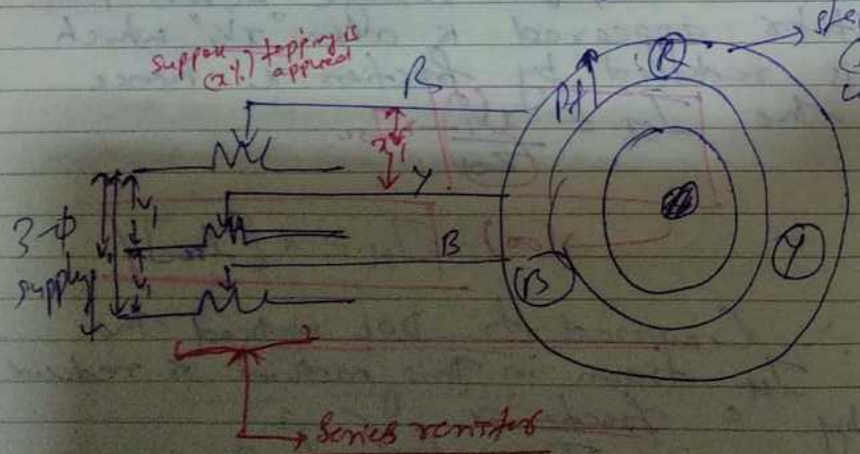
So, starting torque also reduced by fraction of x^2.

So, Compared to DOL method, the starting torque will also reduces by fraction x^2.

Drawback of this method is

→ Due to series reactor, the power factor of I.M reduces which effect the efficiency of I.M.

(3) Series resistor starting method :



So, $I_{st} = \frac{x\%}{Z_{01}} \rightarrow I_{sc}$

So, $I_{st} = x\, I_{sc}$

By $\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_{fl}}\right)^2 s.fl$

(or)

$$\frac{T_{st}}{T_{fl}} = x^2\left(\frac{I_{sc}}{I_{fl}}\right)^2 s.fl$$

If x% tapping is applied on series resistors then the starting current reduces by a fraction x, & the starting torque reduces by a fraction x^2 compared to DOL method.

Advantage

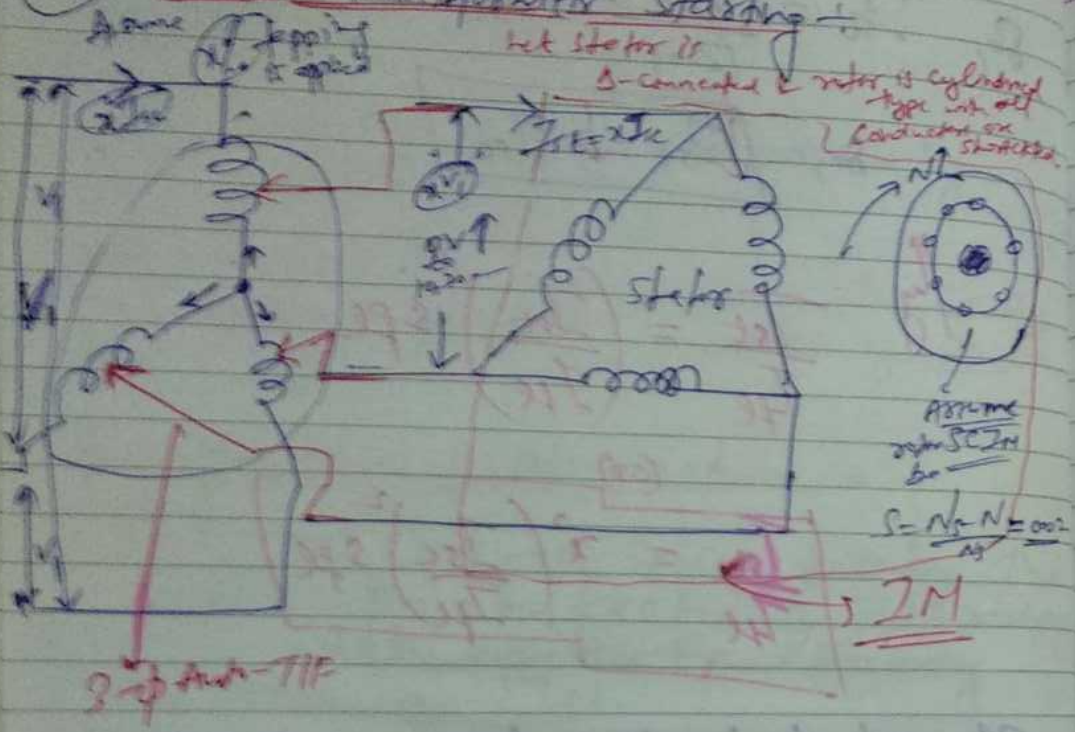
→ The power factor of IM improves by replacing the reactor with resistor.

Disadvantage

→ Due to presence of series resistor the copper loss increasing which reduces the efficiency of motor.

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(4) Auto-Transformer starting :-



If x% tapping is applied on an Auto-TIF then the current drawn by I_M reduces by a fraction x compared to DOL method.

At the same time the current drawn by primary of auto-TIF from supply is reduced by a factor x^2.

Hence $\frac{T_{st}}{T_{fl}} = \left[\frac{I_{st}}{I_{fl}}\right]^2 S_{fl}$

$= \left(\frac{x \cdot I_{sc}}{I_{fl}}\right)^2 S_{fl}$

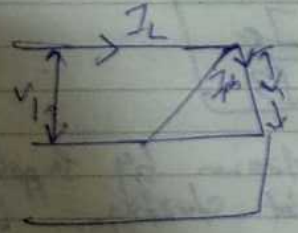
$$\boxed{\frac{T_{st}}{T_{fl}} = x^2 \left(\frac{I_{sc}}{I_{fl}}\right) S_{fl}}$$

Compared to other methods for the same reduction in torque, the current drawn from supply is reduce by fraction x.

Disadvantage is → The size, cost & complexity is more in circuit designing.

(5) Y/Δ starting :-

Assume Δ → Connected stator



Let Z_0 be the phase impedance

$\therefore I_{ph} = \frac{V_L}{Z_0} \rightarrow I_{sc}$

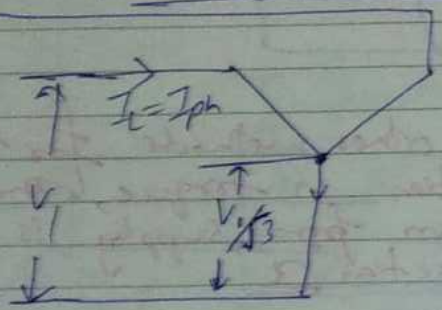
$\therefore I_L = \sqrt{3} I_{ph}$

$\boxed{I_L = \sqrt{3} I_{sc}}$

Here current is less by $\frac{1}{\sqrt{3}}$ at starting & after drawn from supply is also less.

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Assume Y → connected stator



$$\therefore I_{ph} = \frac{V_l/\sqrt{3}}{Z_{ol}}$$

$$\Rightarrow I_{ph} = \frac{1}{\sqrt{3}}\left(\frac{V_l}{Z_{ol}} \rightarrow I_{sc}\right)$$

$$\Rightarrow \boxed{I_{ph} = \frac{I_{sc}}{\sqrt{3}}}$$

$$\therefore \boxed{I_L = I_{ph} = \frac{I_{sc}}{\sqrt{3}}}$$

Hence current drawn by supply is reduced at starting.

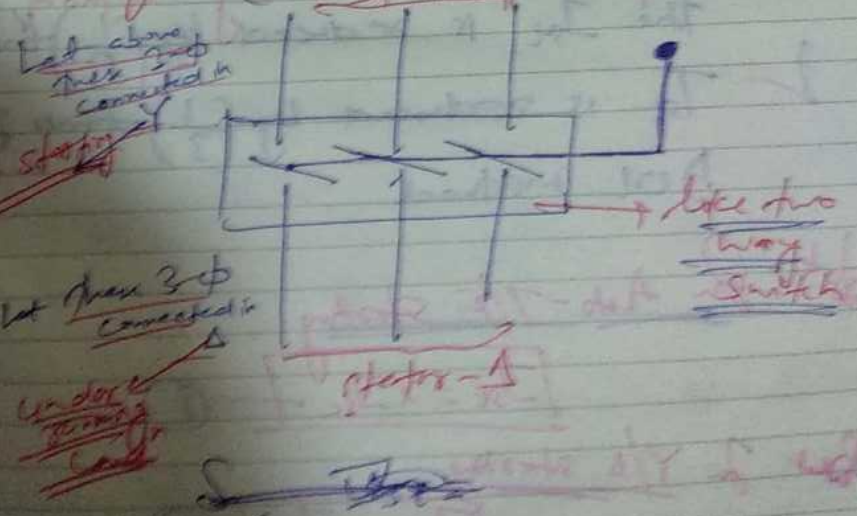
Note: At starting Y connected is preferred but after starting i.e. in rated running condition Δ connection is preferred.

∴ This starting method is possible only when there is a provision of both Star (Y) & delta (Δ) connected winding is present in the IM.

Since, in case of Y connected stator, the starting current drawn is $\left(\frac{1}{\sqrt{3}}\right)$ times of I_{sc} which is limited without any external device.

Hence Y-connected stator is preferred at starting & Δ-connected stator is preferred under running condition.

The change of connection from (Y-Δ) can be made at starting by using (Y/Δ-Switch)



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Y ⟶ $I_{st} = \frac{I_{sc}}{\sqrt{3}}$

& $\frac{T_{st}}{T_{fl}} = \left[\frac{I_{st}}{I_{fl}}\right]^2 \cdot S_{fl}$

$= \left[\frac{I_{sc}/\sqrt{3}}{I_{fl}}\right]^2 \cdot S_{fl}$

$\boxed{\frac{T_{st}}{T_{fl}} = \frac{1}{3}\left[\frac{I_{sc}}{I_{fl}}\right]^2 S_{fl}}$ (Δ-Y)

So, Compare to DOL starting method the I_{st} is reduced by $\left(\frac{1}{\sqrt{3}}\right)$ times

& T_{st} is reduced by $\left(\frac{1}{3}\right)$ times of DOL method

Note: In Auto-T/F starting:

$\boxed{I_{st} = x\, I_{sc}}$ — (1)

But in Y/Δ starting

$\boxed{I_{st} = \frac{I_{sc}}{\sqrt{3}}}$ — (2)

So, comparing eq (1) & (2) we have

$x = \frac{1}{\sqrt{3}} = 0.577$ ⟶ i.e 57.7%

&, At 57.7% Auto-T/F, the starting current drawn by Y/Δ starting & Auto-T/F starting are same

Q The starting current of a 3-ϕ IM is 5 times the rated current while the rated slip is 4%, the ratio of $\frac{T_{st}}{T_{fl}}$ = ?

a) 0.8 b) 0.6 c) 1.2 d) 1

Soln: We know $\boxed{\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_{fl}}\right)^2 S_{fl}}$

Given, $I_{st} = 5 I_{fl}$ & $S_{fl} = 4\% = 0.04$

So $\frac{T_{st}}{T_{fl}} = \left(\frac{5 I_{fl}}{I_{fl}}\right)^2 \times 0.04 = 1$ Ans

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8) The suitable tapping on an auto-T/F when the IM required to start with 40% of full load torque. The short ckt current drawn is 5 times the full load current & full load slip is 0.035 is _____?

Sol: a) 57.7% b) 67.6% c) 66.7%
d) 60%

Given: At start $\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_{fl}}\right)^2 \times s_{fl}$

But in case of Auto-T/F $\frac{T_{st}}{T_{fl}} = x^2 \left[\frac{I_{st}}{I_{fl}}\right]^2 s_{fl}$ — (1)

Given $T_{st} = 40\% \text{ of } T_{fl}$

$I_{st} = 5 I_{fl}$

$s_{fl} = 0.035 = 3.5\%$

So eq (1) become

$0.4 \frac{I_{fl}}{I_{fl}} = x^2 \left(\frac{5 I_{fl}}{I_{fl}}\right)^2 \times 0.035$

$x = 0.676 = 67.6\%$ tapping

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